Journal of Biomedical Informatics 149 (2024) 104571

Contents lists available at ScienceDirect

Journal of Biomedical Informatics

journal homepage: www.elsevier.com/locate/yjbin

Original Research

A data-driven epidemic model with human mobility and vaccination protection for COVID-19 prediction

Ruqi Li [a], Yurong Song [b,*], Hongbo Qu [a], Min Li [b], Guo-Ping Jiang [b]

[a] School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing 210003, China
[b] College of Automation and College of Artificial Intelligence, Nanjing University of Posts and Telecommunications, Nanjing 210023, China

ARTICLE INFO

Keywords:
Data-driven epidemic model
Human mobility
Vaccination protection
Physics-informed neural network

ABSTRACT

Epidemiological models allow for quantifying the dynamic characteristics of large-scale outbreaks. However, capturing detailed and accurate epidemiological information often requires consideration of multiple kinetic mechanisms and parameters. Due to the uncertainty of pandemic evolution, such as pathogen variation, host immune response and changes in mitigation strategies, the parameter evaluation and state prediction of complex epidemiological models are challenging. Here, we develop a data-driven epidemic model with a generalized SEIR mechanistic structure that includes new compartments, human mobility and vaccination protection. To address the issue of model complexity, we embed the epidemiological model dynamics into physics-informed neural networks (PINN), taking the observed series of time instances as direct input of the network to simultaneously infer unknown parameters and unobserved dynamics of the underlying model. Using actual data during the COVID-19 outbreak in Australia, Israel, and Switzerland, our model framework demonstrates satisfactory performance in multi-step ahead predictions compared to several benchmark models. Moreover, our model infers time-varying parameters such as transmission rates, hospitalization ratios, and effective reproduction numbers, as well as calculates the latent period and asymptomatic infection count, which are typically unreported in public data. Finally, we employ the proposed data-driven model to analyze the impact of different mitigation strategies on COVID-19.

1. Introduction

The ongoing emergence of SARS-CoV-2 variants has triggered multiple waves of outbreaks worldwide. On January 30th, 2023, the World Health Organization issued a statement indicating that despite improved global conditions compared to the peak of Omicron transmission a year ago, over 170,000 COVID-19-related fatalities were reported worldwide within the last eight weeks [1]. One of the key reasons that makes controlling this outbreak so arduous is that individuals can transmit the virus during the incubation period, with some remaining asymptomatic throughout the course of the disease [2]. This presents significant challenges to detecting and isolating cases, as these silent infections can easily lead to presymptomatic and asymptomatic transmission. Therefore, it is of utmost importance to closely monitor observed infections, identify potential unobserved high-risk subpopulations, and evaluate their epidemiological and societal impacts.

By leveraging insights from the fields of statistical physics and epidemiology, researchers have developed various mechanism models to assist decision makers in designing effective public health policies. For example, in the early stages of the COVID-19 outbreak in China, Yang et al. [3] integrated population migration into the SEIR model to predict how travel restrictions would impact the spread of the epidemic in Hubei, Guangdong and Zhejiang provinces. Martínez-Fernández et al. [4] compared Verhulst's, Gompertz's, and SIR models from the point of view of their efficiency to describe the behavior of COVID-19 in Spain. Sen et al. [5] employed OLS and IV regression analysis to evaluate the effects of non-pharmaceutical interventions on daily cases in Canadian provinces, and tested the forecasting performance of SIR, GBRT, SARIMA, and VAR models. To control the rapid spread of COVID-19, many countries enforced long-term quarantine and isolation measures, which were analyzed by mathematical modeling methods in [6–8]. Alongside non-pharmaceutical interventions, vaccine development and application has become a significant means of preventing pandemics [9]. Yang et al. [10] designed a SAIVR model incorporating the impact of vaccination to examine the spread dynamics of COVID-19 and discussed the stability conditions of the model. Vivekanandhan et al. [11] proposed a SIR/V model and applied game theory to optimize the vaccination campaign.

* Corresponding author.
E-mail address: songyr@njupt.edu.cn (Y. Song).

https://doi.org/10.1016/j.jbi.2023.104571
Received 13 August 2023; Received in revised form 22 November 2023; Accepted 7 December 2023
Available online 12 December 2023
1532-0464/© 2023 Elsevier Inc. This article is made available under the Elsevier license (http://www.elsevier.com/open-access/userlicense/1.0/).

To understand the qualitative relationship between model parameters and state variables, numerical methods are commonly employed to solve the differential equations describing the system dynamics. In this context, model parameters, such as infection and recovery rates, must be assumed a priori [12]. However, assuming these parameters remain constant over time may limit the accuracy of the model's quantitative predictions. It is essential to adjust these parameters as the pandemic evolves, which in turn presents a challenge in inferring parameters and predicting outcomes.

Data-driven machine learning methods are now being applied to address the complexity and predictability of epidemic dynamics. Ghosh et al. [13] used reservoir computing based echo state network (ESN) to forecast the spread of a disease, and verified the performance of ESN through the synthetic data generated by the SIR model and real COVID-19 case data from various countries. Nikparvar et al. [14] fed mobility as a variable into a multivariate long short-term memory (LSTM) recurrent neural network to predict confirmed cases and deaths in US counties. Meanwhile, to understand the mechanism of contagion on complex networks, Murphy et al. [15] trained a graph neural network on data generated by different contagion dynamics, and demonstrated the approach's utility using real case data from Spain. While these model-free machine learning schemes performed well in terms of predictive fidelity, the underlying coupling relationship and evolution mechanism of multiple states remain largely unclear.

Recent works have shown that hybrid models, which combine physics-based models with machine learning, can strike a balance between interpretability and prediction accuracy [16–18]. The machine learning community has focused extensively on neural network algorithms to achieve this. In [19–21], the contagion mechanism is embedded into a neural network, and the model is trained with available data to generate time-dependent epidemiological parameters. Once the parameters are determined, numerical methods are used to predict future outcomes by solving the differential equations. While these models provide insights into the physical mechanisms underlying state interplay and improve prediction performance to some extent, their loss function during training only focuses on the real-time inference of epidemiological parameters. As such, fully coupling dynamics for direct and efficient state prediction is not yet achieved.

Physics-Informed Neural Networks (PINN) is a machine learning method that combines physical models with deep neural networks to solve complex mathematical problems [22,23]. In order to overcome the limitations of classical hybrid methods, we apply Physics-Informed Neural Networks to a new epidemiological model. In a PINN architecture, a neural network is augmented with physics-based constraints. These constraints are incorporated into the loss function of the network, allowing the network to be trained to not only match the data, but also the governing equations of the physical system. Thus, PINN can learn the relationship between inputs and outputs while maintaining the underlying physical principles of the system. One important advantage of our data-driven scheme is that it allows for the direct inference of epidemiological parameters and unobserved dynamics from data on infected cases, hospitalizations, and deaths using a separate neural network. Additionally, the flexibility of the neural network architecture allows for the integration of both epidemiological data and additional physical measurements, such as vaccination and population movement, to improve the accuracy and interpretability of the model.

To restore normal life and production, countries have continuously optimized epidemic prevention and treatment measures. Notably, individuals with asymptomatic or mild infections are no longer required to undergo hospitalization. Moreover, even if individuals contract the virus, they can continue their daily activities without mandatory quarantine. With Omicron becoming the dominant strain, the asymptomatic infection of novel coronavirus has gradually increased and the death rate caused by the virus has decreased [24]. Some asymptomatic infected patients do not develop clinical symptoms due to their own immunity, while others may progress to symptomatic confirmed cases. Given our previous understanding of the transmission mechanism and intervention strategies of COVID-19, we designed a more detailed $SEI^{S}I^{a}HDR$ state-dynamics model. In addition to accounting for the transmission potential of asymptomatic infections, the proposed model differs from the traditional SEIR model in that we take into account the impact of vaccine uptake and human mobility on disease transmission, while also introducing additional compartments to better capture the complexity of the real-world situation.

In this paper, we show how the physics-informed neural network (PINN), which is commonly used in structural learning, can be applied to model infectious disease dynamics. Our contributions are three-fold: Firstly, we develop a data-driven epidemiological model that integrates mechanistic model with deep learning. Secondly, we propose a new $SEI^{S}I^{a}HDR$ model that considers human activities and vaccination protection, and validate the effectiveness of our approach through the pandemic data from Australia, Israel, and Switzerland. Thirdly, we demonstrate the model's capability to infer time-varying parameters and unobserved states, as well as make predictions under different scenarios in the future. The following steps show the implementation of our modeling framework:

(I) Acquiring case data and quantifying significant interventions that may influence the spread of the epidemic, followed by preprocessing the data.
(II) Developing an epidemiological mechanistic model that incorporates the data and interventions from step (I).
(III) Utilizing physics-informed neural networks to evaluate time-varying parameters and state variables of the model. Incorporating prior knowledge to optimize the model's output.
(IV) Applying the trained model to forecast the future progression of the epidemic.
(V) Investigating how changes in interventions affect the dynamic propagation of the epidemic.

Summary	Description
Problem:	The pursuit of both accuracy and interpretability in prediction models is challenging due to the uncertainty inherent in the evolution of large-scale epidemics.
What is Already Known:	Despite there have been many prior studies on epidemic prediction, the underlying physical propagation mechanisms have received little attention in the quest for quantitative accuracy.
What This Paper Adds:	We develop a data-driven epidemic model with a generalized SEIR mechanisms including new compartments, human mobility, and vaccination protection. Dynamic mechanisms are embedded into physics-informed neural networks to simultaneously infer unknown parameters and unobserved dynamics. The results show the predictability, robustness, and interpretability of the model.

2. Methods

2.1. Dynamic model

Various studies have found that individuals with asymptomatic and symptomatic SARS-CoV-2 infections have a comparable viral load, supporting the idea of similar transmission potential [25]. Many countries adopt home care policies for asymptomatic or mildly symptomatic infections, while severe cases receive hospital treatment. To account for transmissions from pre-symptomatic, asymptomatic and symptomatic

individuals, as well as treatment options for different infection states, we modify the classic SEIR model by introducing asymptomatic, hospitalized, and disease-related death compartments.

Moreover, recent studies have underscored the potential significance of monitoring human mobility patterns in facilitating epidemiological analysis [26–28]. Unlike previous research on the impact of overall mobility on the pandemic, our investigation utilizes human activity data from the Google Mobility Reports to analyze how distinct categories of activities affect the transmission of COVID-19. Specifically, the residential category offered in the Mobility Reports measures the reduction in contact between susceptible and infected individuals, which is believed to suppress the spread of the virus. In contrast, outdoor activities in settings such as retail & recreation, parks, grocery & pharmacy stores, workplaces, and transit stations promote more frequent interaction among individuals as leaving shelters, which may increase the risk of secondary infections.

In addition to non-pharmaceutical interventions mentioned above, vaccination campaigns implemented worldwide are also crucial in safeguarding the uninfected population [9,29]. Therefore, we introduce a transient immunity factor c into the model, which implies that after effective vaccination, the population will not be infected at the current time step with probability c. Since the vaccine efficacy is not 100%, we calculate the number of effectively vaccinated individuals using a weighted sum of the first and second dose administered.

Based on the insights gained from studying the COVID-19 pandemic, we develop an SEI^{s}I^{a}HDR model that incorporates human activity and vaccine protection to more accurately describe the progression of the epidemic. The proposed model consists of eight states: susceptible (S), exposed (E), symptomatic infected (I^s), asymptomatic infected (I^a), hospitalized (H), disease-related death (D) and recovered/removed (R); see 1. Fig. 1 illustrates the outline of the SEI^{s}I^{a}HDR compartmental model to explain the transition between different epidemiological states.

The mean-field equation system of our model is the following:

$$\begin{cases} \frac{\mathrm{d}S(t)}{\mathrm{d}t} = -\frac{\beta\left[I^c(t)+\kappa(E(t)+I^a(t))\right]}{N}S(t) - \frac{cV}{N}S(t) - \Delta_{in}S(t) + \Delta_{out}R(t) \\ \frac{\mathrm{d}E(t)}{\mathrm{d}t} = \frac{\beta\left[I^c(t)+\kappa(E(t)+I^a(t))\right]}{N}S(t) - \alpha E(t) \\ \frac{\mathrm{d}I^s(t)}{\mathrm{d}t} = \rho\alpha E(t) - d_s I^s(t) \\ \frac{\mathrm{d}I^a(t)}{\mathrm{d}t} = (1-\rho)\alpha E(t) - d_a I^a(t) \\ \frac{\mathrm{d}H(t)}{\mathrm{d}t} = \omega d_s I^s(t) - d_H H(t) \\ \frac{\mathrm{d}D(t)}{\mathrm{d}t} = \mu d_H H(t) \\ \frac{\mathrm{d}R(t)}{\mathrm{d}t} = (1-\mu)d_H H(t) + (1-\omega)d_s I^s(t) + d_a I^a(t) + \left(\frac{cV}{N} + \Delta_{in}\right)S(t) - \Delta_{out}R(t) \end{cases} \tag{1}$$

The initial values of the observed states I^s, H and D, denoted as $I^s(0)$, $H(0)$ and $D(0)$, depend on the initial values of the collected dataset. For the initial values of the unobserved states, we set $E(0) = I^s(0)$, $I^a(t) = 0$, $R(0) = 0$, and $S(0) = N - E(0) - I^s(0) - I^a(0) - H(0) - D(0) - R(0)$. The population size N in a region can be obtained from the United Nations Demographic Yearbook.

Susceptible individuals are infected by symptomatic individuals with probability β, infected by pre-symptomatic and asymptomatic individuals with probability $\kappa\beta$. After an incubation period $1/\alpha$ days, exposed individuals develop into symptomatic infections with probability ρ, and develop into asymptomatic infections with probability $1-\rho$. Symptomatic individuals are hospitalized after an infectious period $1/d_s$ days. It is assumed that hospital cases will not infect others under strict management [30]. The inpatients move to recovered compartment with probability $1-\mu$ after d_H days of treatment, and the remainder died. Asymptomatic infected individuals recovered after the infection period $1/d_a$ days. The effective or control reproduction number of the model is given by $R_c = \beta\left(\frac{\rho}{d_s} + \frac{\kappa(1-\rho)}{d_a} + \frac{\kappa}{\alpha}\right)$. For a list of parameters and variables, see Table 2.

2.2. Physics-informed neural network architecture

To understand the transient behavior of epidemiological parameters and predict future epidemic trends, we employ physics-informed neural networks (PINNs) and detail their application to the SEI^{s}I^{a}HDR model. As shown in Fig. 2, PINN consists of input layers, fully connected hidden layers with nonlinear activation functions, and output layers. The neuron state of each layer is updated through the weight matrix w and the bias b. We construct independent neural networks to infer dynamic parameter values, observed states, and unobserved states at each discrete time t_n.

For the evaluation of time-dependent parameters, the proposed feed-forward neural network structure consists of a hidden layer containing 5 neurons. For the evaluation of state quantities, we use 5 hidden layers with 20 neurons in each layer. The neurons use the tanh kernel as the network activation function. The output layer is fully connected to the hidden layer through neurons of the desired output dimension, where the sigmoid function is used to activate the neurons.

Model (1) can be rewritten as a general representation

$$\frac{du}{dt} + \mathcal{N}[u(t), \chi] = 0, \tag{2}$$

where $u(t)$ is a placeholder for states $\{S(t), E(t), \ldots, R(t)\}$, $\mathcal{N}$ is a nonlinear operator, and χ is a fixed/fitted set of model parameters. Within the PINN framework, we utilize the available data $u(t)$ to train the state approximation $u_{NN}(t)$ and the time-varying parameter approximation $\chi_{NN}(t)$ using separate deep neural networks. By optimizing the weights and biases of each network, we aim to approximate the solution of the differential equation. Therefore, we have $u(t) \approx u_{NN}(t)$, $\chi(t) \approx \chi_{NN}(t)$.

Using Eq. (2), we define the residual $f(t)$ of the equation during the training process as follows:

$$f(t) := \frac{du_{NN}}{dt} + \mathcal{N}\left[u_{NN}(t), \chi\right], \tag{3}$$

this residual measures the deviation of the model's approximate solution u_{NN}. Ideally, u_{NN} would be considered as an exact solution when the residual is equal to zero. To account for both the mismatch of training points and the ODE residual, we define the loss function of the Physics-Informed Neural Network as:

$$L(\beta(t), \omega(t), \mu(t)) = MSE_u + MSE_f, \tag{4}$$

where

$$MSE_u = \frac{1}{N_u}\sum_{i=1}^{N_u}\left|u_{NN}\left(t_u^i\right) - u_D\left(t_u^i\right)\right|^2, \tag{5}$$

and

$$MSE_f = \frac{1}{N_f}\sum_{i=1}^{N_f}\left|f\left(t_f^i\right)\right|^2. \tag{6}$$

In our experiment, we utilize the TensorFlow learning framework based on python (version 3.7) to train the network's weight w and bias b in order to solve the optimization problem associated with the PINN. The optimization process is carried out over a pre-set training cycle N, with u_D representing the available data. TensorFlow provides the necessary tools and functionalities for efficiently optimizing the network parameters and finding the optimal solution for the PINN problem.

3. Result

3.1. Data source and quality evaluation metrics

We validate the proposed model using COVID-19 epidemic data from three countries, namely Australia, Israel, and Switzerland, which present different pandemic situations. To avoid lag and outliers in case reporting, the dataset is preprocessed with a seven-day moving average window strategy.

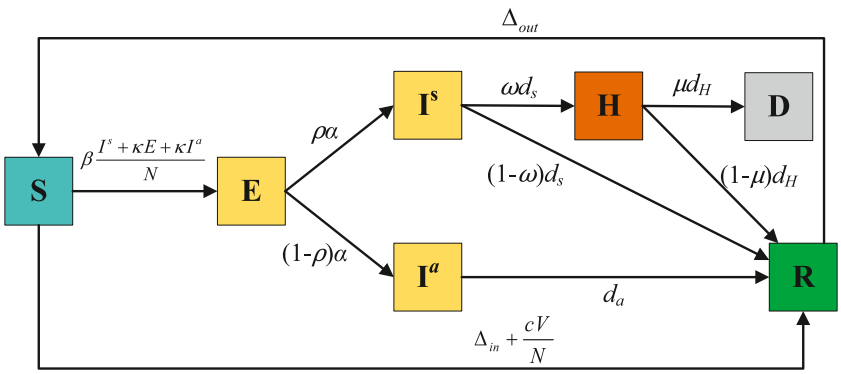

Fig. 1. Compartment diagram of the SEI^{s}I^{a}HDR model.

Table 1
Notations and initial values for the epidemiological model.

Notation	Description	Initial value/Note
N	Population size in a region	Assume constant total population [31]
$S(t)$	Number of susceptible individuals at time t	$S(0) = N - E(0) - I^s(0) - I^a(0) - H(0) - D(0) - R(0)$
$E(t)$	Number of exposed individuals at time t	$E(0) = I^s(0)$
$I^s(t)$	Number of symptomatic individuals at time t	$I^s(0)$ from data [32]
$I^a(t)$	Number of asymptomatic individuals at time t	$I^a(t) = 0$
$H(t)$	Number of hospitalized individuals at time t	$H(0)$ from data
$D(t)$	Number of disease-related death at time t	$D(0)$ from data
$R(t)$	Number of recovered/removed individuals at time t	$R(0) = 0$
$V(t)$	Number of effectively vaccinated individuals at time t	$V(0)$ from data
$\Delta_{in}(t)$	Residential activity index at time t	$\Delta_{in}(0)$ from data
$\Delta_{out}(t)$	Outdoor activity index at time t	$\Delta_{out}(0)$ from data

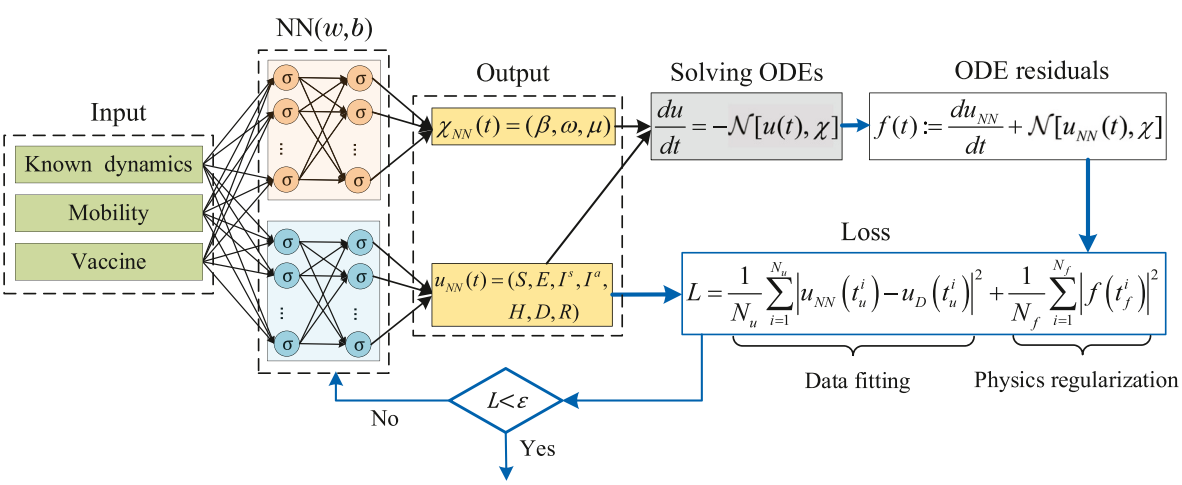

Fig. 2. Schematic diagram of the physics-informed neural network for SEI^{s}I^{a}HDR model.

Since the Pfizer vaccine is the preferred vaccine recommended by these three countries, the model utilizes effectiveness parameters specific to the Pfizer vaccine. The vaccine efficacy is 52% for only one dose and 95% for both doses [40]. To assess the vaccine efficacy of vaccinated populations, we use the weighted sum of the first and second doses: $v = 0.52v_1 + 0.95v_2$, where v_1 is the proportion of individuals vaccinated with the first dose, and v_2 is the proportion of individuals vaccinated with the second dose. We multiply the daily vaccination data by the vaccine efficacy to generate the effective vaccination population, and then use the following Gaussian function to approximate the daily number of effective vaccination individuals:

$$V(t) = v\exp(-((t-\tau)/\xi)^2) + \sigma. \tag{7}$$

Here, the parameters τ, ξ, σ are determined by the collected vaccination data (see Fig. 3). Real case data and vaccine administration data can be downloaded from https://ourworldindata.org/coronavirus. Furthermore, we incorporate the Google Community Mobility Report into the proposed model. In response to the COVID-19 pandemic, countries around the world developed a series of mitigation strategies, such as home quarantine, cancellation of public gatherings, and restrictions on public transportation. These Community Mobility Reports provide insights into how people's social behaviors have been changing as a result of mitigation strategies. This dataset from Google measures daily visitor numbers to different activity categories in specific countries or regions, and gives the changes in movement trends compared to pre-pandemic baselines. These activities include retail & recreation,

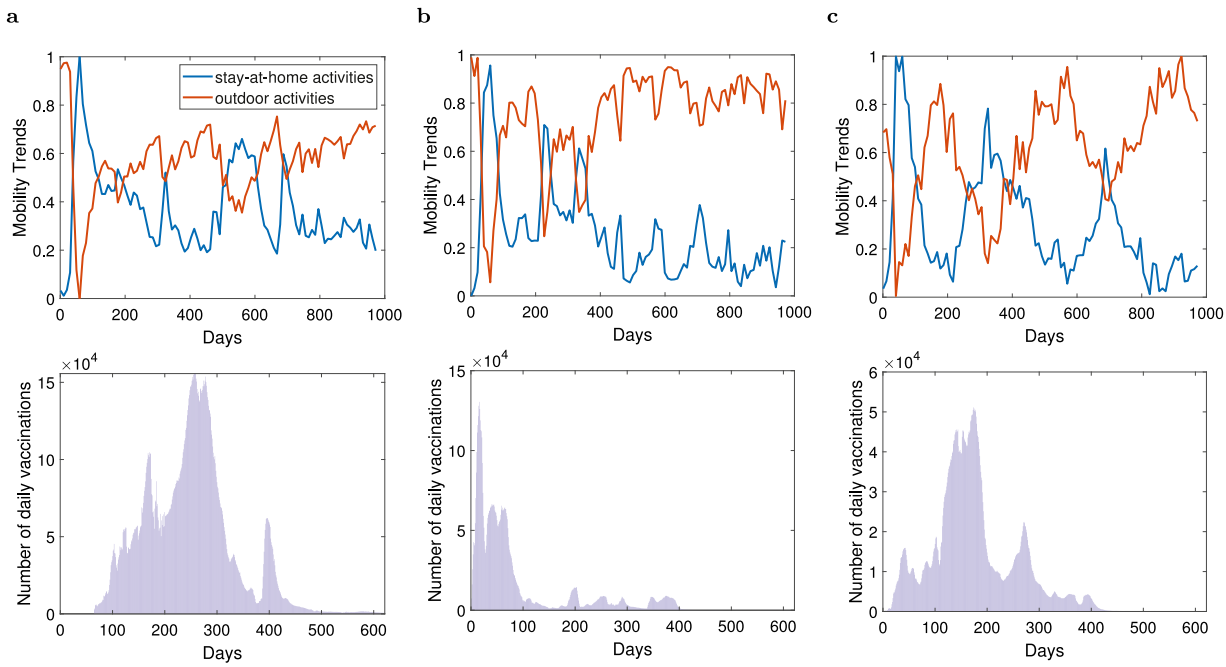

Fig. 3. Changes in population mobility and vaccination in (a) Australia, (b) Israel and (c) Switzerland since the beginning of the pandemic.

Table 2
Parameters for the epidemiological model.

Parameter	Description	Value/Note
β	Transmission rate	Fitted
ω	Hospitalization ratio	Fitted
μ	Death from hospital ratio	Fitted
κ	Infectivity ratio of asymptomatic to symptomatic	0.55 [33]
ρ	Proportion of symptomatic infections	0.75 [34]
$1/c$	Vaccine Onset Period	14 days [35]
$1/\alpha$	Incubation period	3.4 days [36]
$1/d_s$	Infectious period for symptomatic	2.9 days [37]
$1/d_a$	Infectious period for asymptomatic	6 days [38]
$1/d_H$	Hospitalization days	7 days [39]

groceries & pharmacies, parks, transit stations, workplaces, and residential. The relevant data can be downloaded from https://www.google.com/covid19/mobility/.

We create two time series from the Google Community Mobility Report. We amalgamate data from retail & recreation, groceries & pharmacies, parks, transit stations, workplaces, as these locations are considered potential contributors to the secondary transmission of COVID-19. On the other hand, data for residential category describe stay-at-home activities, which are considered to prevent COVID-19 pneumonia. As shown in Fig. 3, we normalize the above two time series, where $\Delta_{out}(t)$ represents data for outdoor activities, and $\Delta_{in}(t)$ represents data for stay-at-home activities. We approximate these two types of activity indices using the following polynomial regression function:

$$\Delta(t) = w_0 + w_1 t + w_2 t^2 + \cdots + w_M t^M, \tag{8}$$

where the parameters w_i are determined by the mobility data, and M is the order of the polynomial. Here, we set $M = 6$.

We input daily new infections I^s_{new}, hospitalizations H_{new} and deaths D_{new} data into the kinetic model (1) via calculating the following system of equations:

$$\begin{cases} I^s_{new}(t) = I^s_{sum}(t) - I^s_{sum}(t-1) \\ H_{new}(t) = H_{sum}(t) - H_{sum}(t-1) \\ D_{new}(t) = D_{sum}(t) - D_{sum}(t-1) \end{cases} \tag{9}$$

where

$$\begin{cases} \dfrac{dI^s_{sum}}{dt} = \rho\alpha E \\ \dfrac{dH_{sum}}{dt} = \omega d_s I^s \\ \dfrac{dD_{sum}}{dt} = \mu d_H H. \end{cases} \tag{10}$$

The quality of the predictions is assessed using the Mean Absolute Percentage Error (MAPE) and Normalized Root Mean Square Error (NRMSE), their mathematical expressions are as follows:

$$\mathrm{MAPE}\left(Y_i, \tilde{Y}_i\right) = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \tilde{Y}_i}{Y_i}\right| \times 100, \tag{11}$$

and

$$\mathrm{NRMSE}\left(Y_i, \tilde{Y}_i\right) = \frac{1}{n}\frac{\sqrt{\sum_{i=1}^{n}\left(Y_i - \tilde{Y}_i\right)^2}}{\bar{Y}}, \tag{12}$$

where Y_i is the observed values, $\tilde{Y}_i$ is the predicted values, n is the number of sample points, and $\bar{Y}$ represents the average value of the observed data.

3.2. Evaluation of PINN based on $\mathrm{SEI^sI^aHDR}$ model

We compare the performance of our method with three data-driven methods that have been utilized to predict the evolution of COVID-19 in multiple countries: LSTM [14,41], logistic growth model [42], and an equivalent SIRX mechanistic model driven by a simple transmission mechanism [6]. The datasets for Australia and Israel start from December 2021, while the dataset for Switzerland starts from July 2021, encompassing a new wave of transmission processes triggered by the Delta and Omicron variants.

Fig. 4 illustrates the comparison between the observations and predictions of our PINN model (red) and other models (Logistic model in purple, LSTM in blue, SIRX in black) across three datasets. During the 50-day training process, all four data-driven methods can effectively reproduce the growth of cases. Table 3 demonstrates the comparison of the four methods in terms of MAPE and NRMSE for the cumulative confirmed cases during a 10-day forecasting period. We can verify that

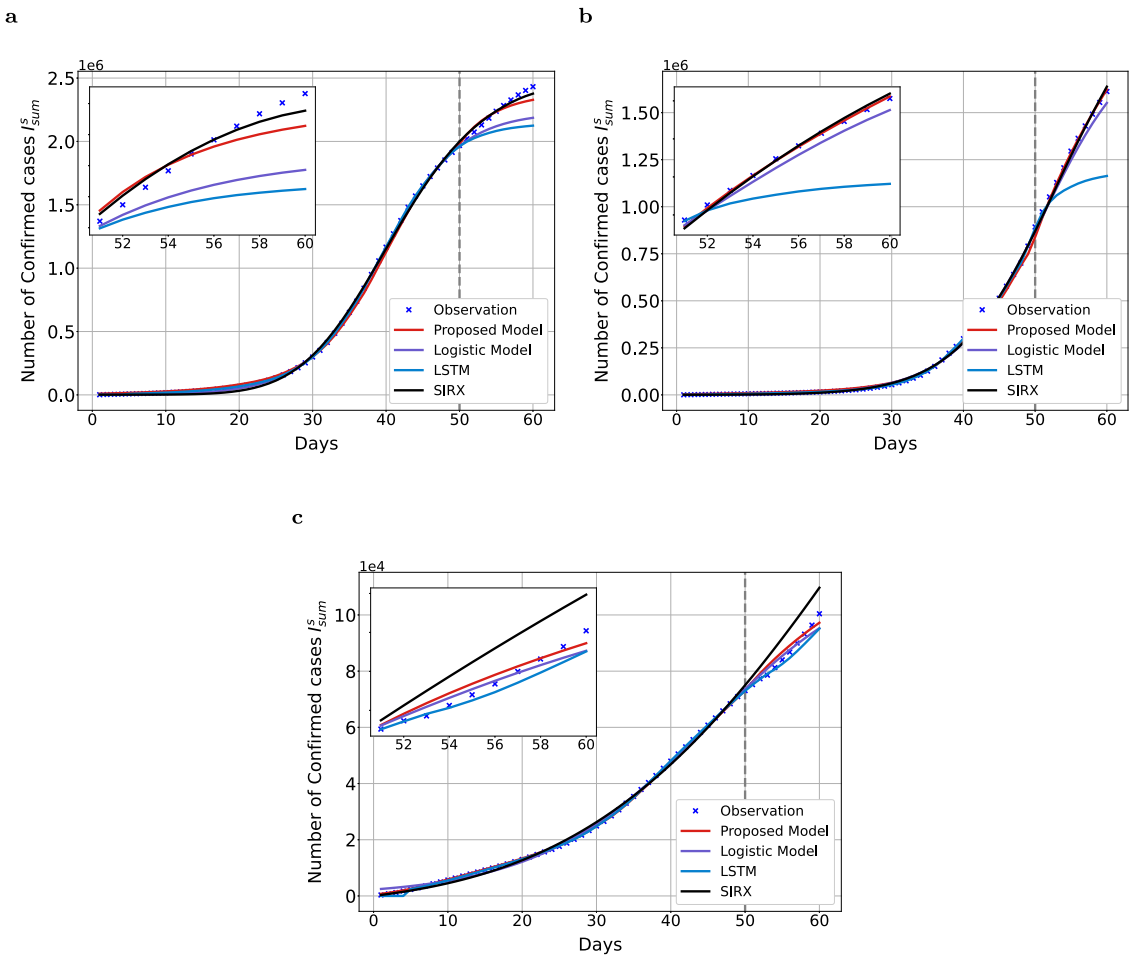

Fig. 4. Comparison between the observations and the predictions for our PINN model (red) and other models (Logistic model in purple, LSTM in blue, SIRX in black) in three datasets, namely (a) Australia, (b) Israel, and (c) Switzerland. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
MAPE and NRMSE for a 10-day forecasting period plotted in Fig. 4.

Country	Metric	Proposed model	Logistic model	LSTM	SIRX
Australia	MAPE	1.977	5.601	7.251	1.257
	NRMSE	0.024	0.067	0.086	0.014
Israel	MAPE	0.789	3.211	14.849	1.423
	NRMSE	0.010	0.034	0.199	0.015
Switzerland	MAPE	2.368	2.095	2.228	8.341
	NRMSE	0.026	0.026	0.031	0.088

our model closely matches or even surpasses the accuracy of other data-driven methods, as the calculated highest MAPE is less than 3%, indicating very small prediction errors. Compared to the model-free LSTM approach, the logistic model and SIRX model are better suited for learning dynamic structural problems, even though the logistic model is designed solely for population growth and the SIRX model incorporates an additional detection state X to the classic SIR model. Furthermore, our SEI^{s}I^{a}HDR-based PINN model is capable of simultaneously learning more detailed dynamics, including the inference of asymptomatic, hospitalized, and death compartments, providing richer epidemiological parameter information. Additionally, our model allows for handling human activities and immune movements, including adjusting vaccination parameters and activity categories to understand their roles in disease dynamics, as detailed next.

As shown in Fig. 5, the three time-dependent parameters (β, ω, μ) for Australia, Israel and Switzerland generated by the PINN algorithm. From this, we can calculate the control reproduction number R_c, which estimates the risk of infectious disease spreading in the region. If R_c is greater than 1, it indicates that the disease will spread quickly and potentially cause a large-scale epidemic, whereas if R_c is less than 1, the disease will spread more slowly and gradually fade out [6]. Fig. 6 illustrates the variations in the control reproduction numbers R_c in Australia, Israel, and Switzerland. The R_c values in Australia and Israel initially decreased, followed by a gradual increase and stabilization, while Switzerland experienced a period of stability at low values during the observation period, followed by a sustained rise. It is worth noting that the final estimated reproduction numbers in all three countries are greater than 1, indicating that the epidemic continues to spread.

Fig. 7 presents a comparison between the observed and predicted data sets, which include new cases, hospitalizations, and deaths. By examining the daily new case trends in Fig. 7, it can be observed that the first wave of the outbreak, which started in December 2021 in Australia and Israel, was significantly higher than subsequent waves, and Switzerland also experienced a notable outbreak during the same period. This phenomenon may be attributed to the transition from the Delta variant to the Omicron variant in December [43]. In Fig. 8, We infer unobserved dynamics from the observed states, which displays the current number of susceptible, exposed, asymptomatic and removed individuals.

By employing the estimated parameters, we present out-of-sample dynamic predictions under three mitigation measures, as shown in Fig. 9. We maintain the pre-prediction vaccination rate and population mobility intensity as the baseline scenario. Then, we implement a

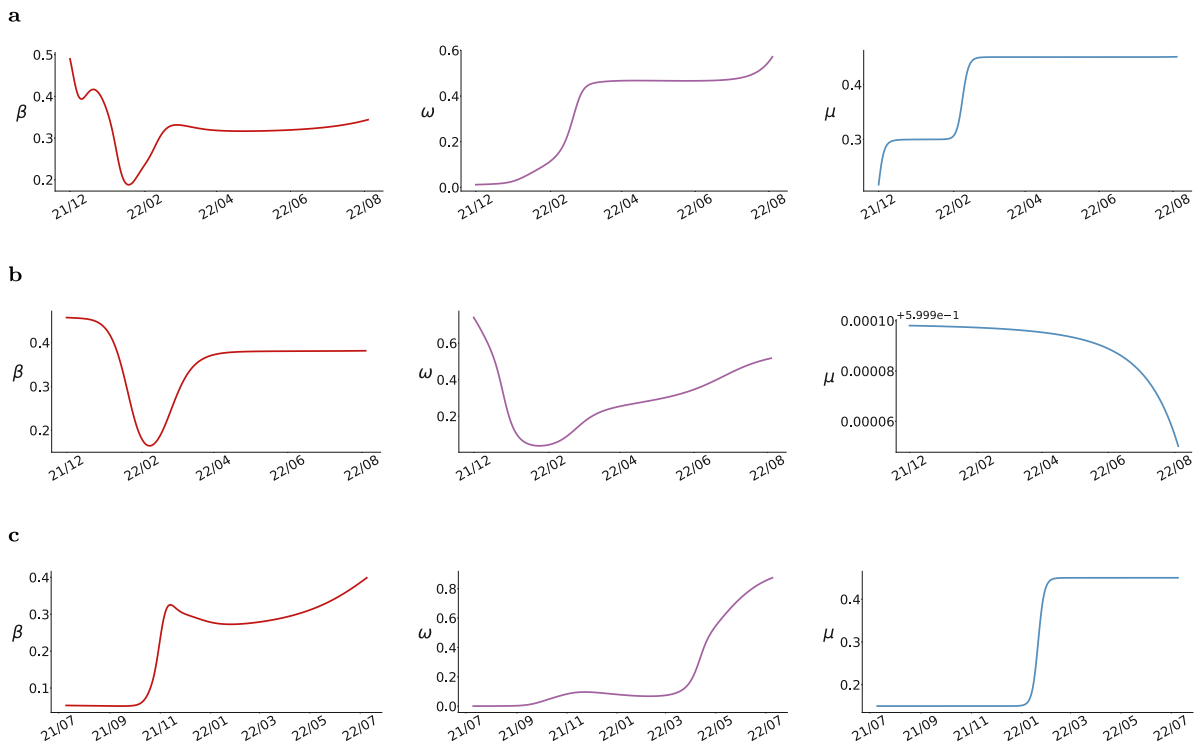

Fig. 5. Inference of time-dependent parameters (β, ω, μ) for (a) Australia, (b) Israel and (c) Switzerland.

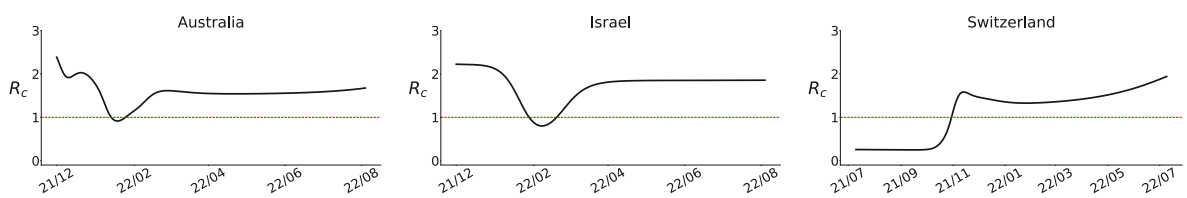

Fig. 6. Estimated control reproduction number (a) Australia, (b) Israel and (c) Switzerland.

relaxation control strategy (reducing the vaccination rate by 10% and increasing population mobility by 10%) and a strict control strategy (increasing the vaccination rate by 10% and decreasing population mobility by 10%) for comparison. It can be observed that the actual data gravitates towards the projected trend of relaxation control. The predictive accuracy under the relaxation strategy is quantified using the mean absolute percentage error (MAPE) as presented in Table 4. The table reveals that the in-sample forecast accuracy for daily infection and hospitalization numbers exceeds 90% in all three countries. However, the prediction accuracy for death cases is relatively diminished owing to the sparse and dispersed characteristics of the daily death data. Similar results are obtained for the out-of-sample forecast within a 14-day period. Overall, these data results demonstrate the effectiveness of the proposed model in short-to-medium term predictions.

Table 4
MAPE for in-sample forecasting period plotted in Fig. 7 and out-of-sample forecasting period plotted in Fig. 9.

Country	Variable	In-sample	Out-of-sample			
			7 days	14 days	21 days	28 days
Australia	I^s_{new}	3.547	7.789	13.898	21.633	24.573
	H_{new}	1.810	1.668	1.313	2.365	4.780
	D_{new}	15.827	10.213	9.097	7.955	8.111
Israel	I^s_{new}	7.237	3.005	6.305	6.704	7.282
	H_{new}	2.767	1.058	2.740	6.068	10.229
	D_{new}	42.829	22.983	43.686	41.521	44.570
Switzerland	I^s_{new}	6.846	3.478	6.046	15.377	28.898
	H_{new}	2.902	2.645	4.674	5.602	4.943
	D_{new}	39.589	14.801	13.962	22.660	20.323

4. Discussion and conclusion

In this work, we propose a data and physics-driven algorithm that combines deep learning and compartmental models for effectively assessing the COVID-19 pandemic. By integrating the generalized SEIR model with PINN, we utilize daily reported data on symptomatic infections, hospitalizations, and deaths to uncover the transmission patterns in Australia, Israel, and Switzerland, and infer the typically unreported incubation period and asymptomatic population in public data. By incorporating population mobility data and vaccination data into parameter estimation process, we are able to capture the impact of individual behavior and policy decisions on the dynamics of the pandemic.

Through comparisons with three data-driven benchmark methods, our model exhibits similar or even superior prediction accuracy. While classical mechanistic models can provide qualitative information through numerical experiments, they often lack the quantitative prior knowledge required for accurate predictions. This is particularly true if time-varying parameters are not considered, as these models may only provide estimates of the early trends in epidemic spread. The subsequent development of the epidemic becomes difficult to predict due to factors such as changing intervention measures and viral characteristics during a pandemic. To address this, we utilize data from different scenarios and incorporate the structure of the mechanistic model into

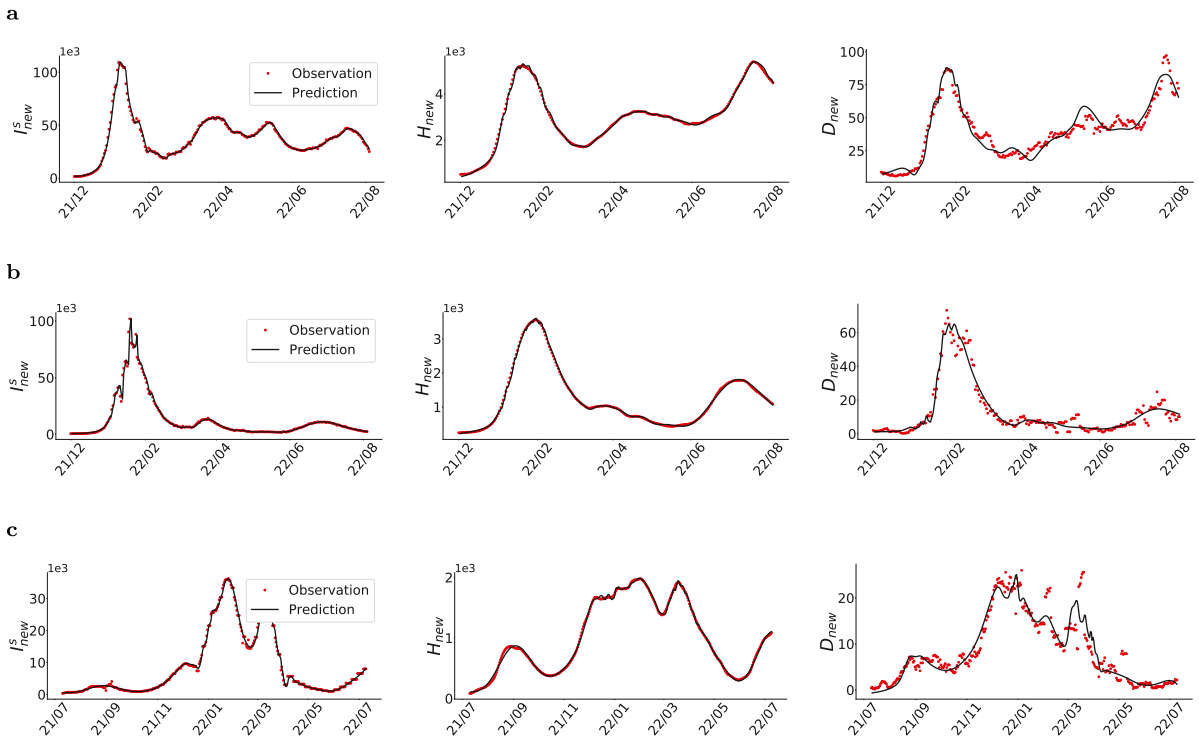

Fig. 7. Fitting available new case data for (a) Australia, (b) Israel and (c) Switzerland.

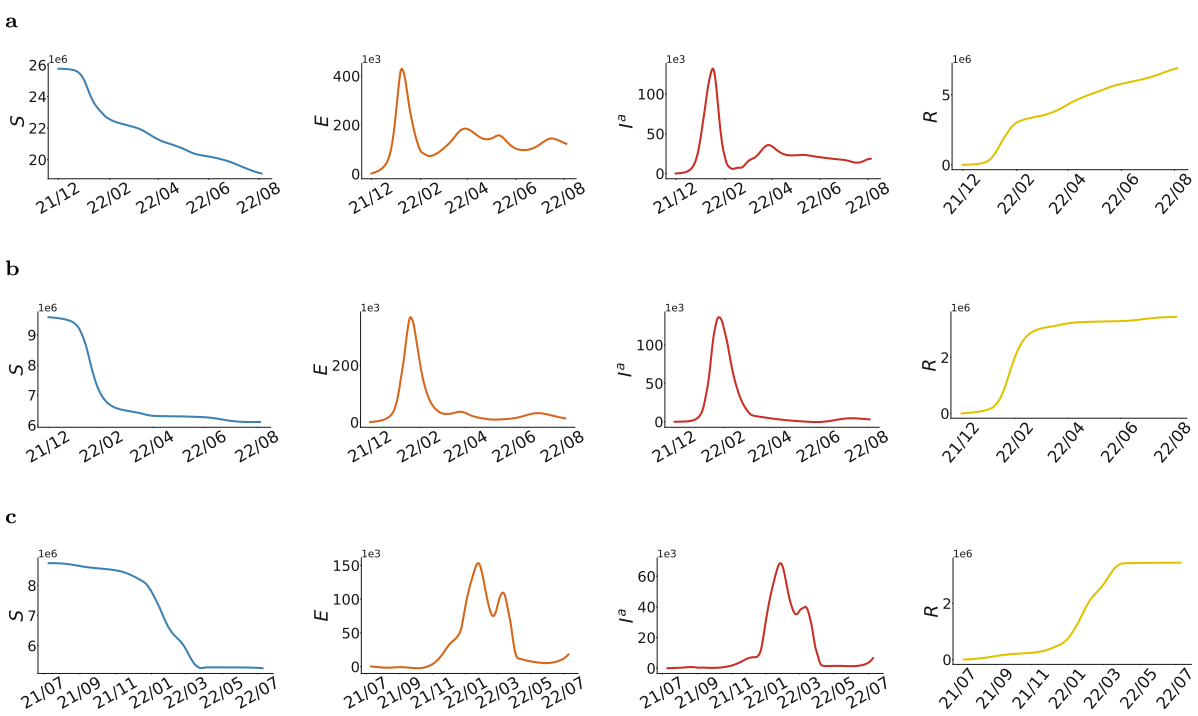

Fig. 8. Inference of unobserved dynamics for (a) Australia, (b) Israel and (c) Switzerland.

neural networks for non-linear representation learning. Through this approach, we are able to estimate time-dependent parameters (such as the transmission rate, hospitalization ratio to death ratio) over the long term and derive the effective reproductive number as an epidemiological measure. Our model allows us to monitor multiple waves of the epidemic and provides detailed insights into the changing trends of both observable populations (daily cases, hospitalizations, and deaths) and unobservable populations (such as the incubation period and asymptomatic individuals). As a result, this approach also enhances

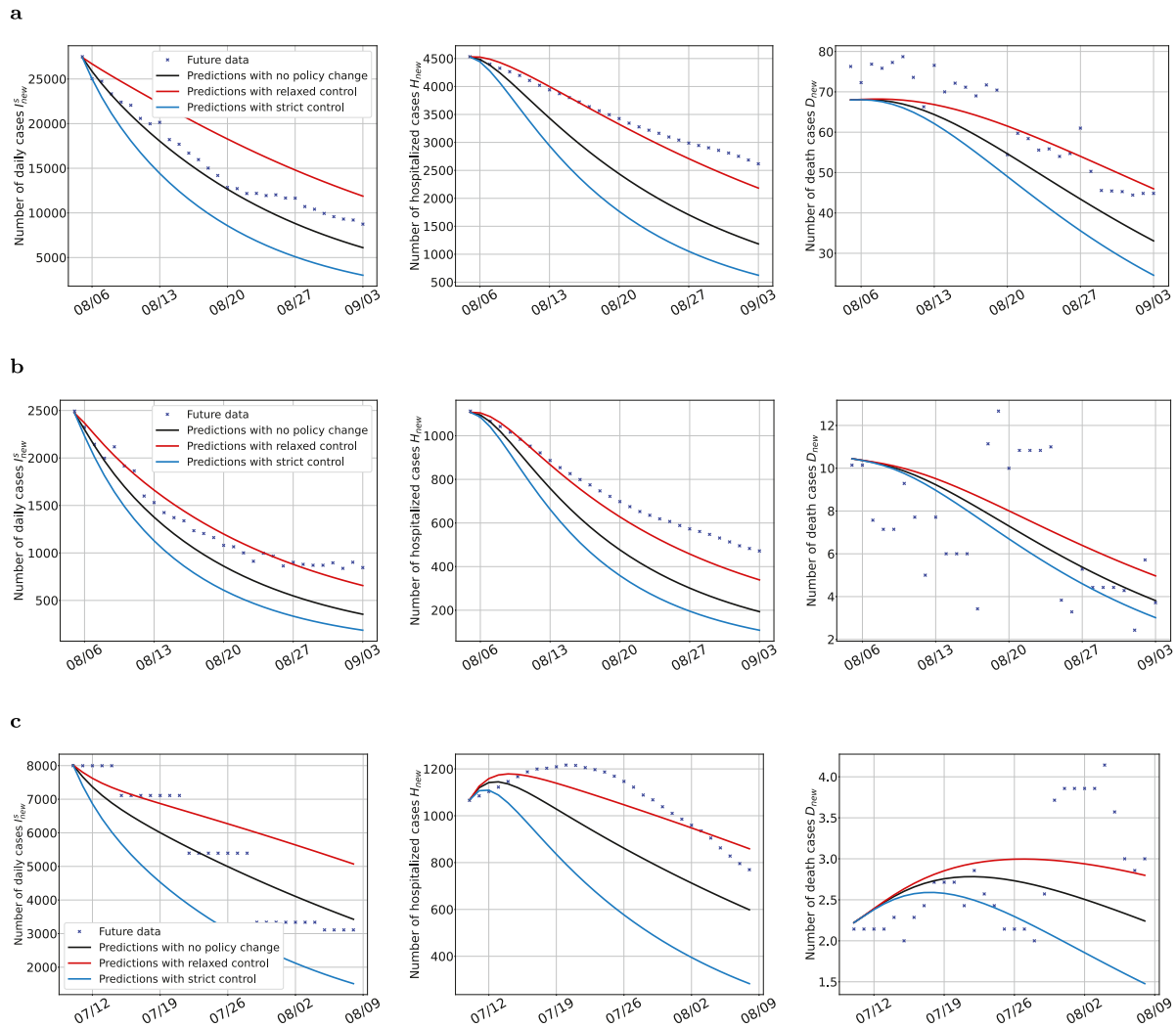

Fig. 9. Forecasting the number of COVID-19 infections for (a) Australia, (b) Israel and (c) Switzerland under different interventions.

the interpretability of predictions, overcoming the black-box nature of pure deep learning.

Given the assumptions and identifiability of the model structure, the current work has some limitations. (I) We have overlooked the impact of variant strains on vaccine efficacy and the issue of vaccine durability. Although we have not explicitly incorporated the decay of vaccine efficacy into the model, we have simplified the representation of transient immune factors that influence the strain's escape ability. (II) Considering the low proportion of hospital–acquired infections, we assume that individuals hospitalized do not infect others. (III) We have only considered one-dimensional time series for model prediction.

In future studies, we aim to address these limitations and improve our methodology to enhance the accuracy and interpretability of the model. By tackling these issues, we seek to further refine our understanding of the effects of variant strains on vaccine effectiveness, explore the influence of acquired infections in specific settings (such as hospitals and schools), and advance the identifiability of spatio-temporal epidemic models to predict disease spread in multiple dimensions.

CRediT authorship contribution statement

Ruqi Li: Data curation, Investigation, Methodology, Writing – original draft. **Yurong Song:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Hongbo Qu:** Resources, Software, Writing – review & editing. **Min Li:** Formal analysis, Writing – review & editing. **Guo-Ping Jiang:** Funding acquisition, Methodology, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported partially by the National Natural Science Foundation of China (Grant Nos. 62373197, 61672298, 61873326), Major Project of Philosophy and Social Science Research in Colleges and Universities in Jiangsu Province, China (Grant No. 2018SJZDI142), Postgraduate Research & Practice Innovation Program of Jiangsu Province, China (Grant No. KYCX22_1018).

References

[1] World Health Organization, Statement on the fourteenth meeting of the International Health Regulations (2005) Emergency Committee regarding the coronavirus disease (COVID-19) pandemic, Available online: https://www.who.int/zh/news.

[2] D.P. Oran, E.J. Topol, Prevalence of asymptomatic SARS-CoV-2 infection: a narrative review, Ann. Intern. Med. 173 (5) (2020) 362–367, http://dx.doi.org/10.7326/M20-3012.

[3] Z. Yang, Z. Zeng, K. Wang, et al., Modified SEIR and AI prediction of the epidemics trend of COVID-19 in China under public health interventions, J. Thorac. Dis. 12 (3) (2020) 165–174, http://dx.doi.org/10.21037/jtd.2020.02.64.

[4] P. Martínez-Fernández, Z. Fernández-Muñiz, A. Cernea, et al., Three Mathematical Models for COVID-19 Prediction, Mathematics 11 (3) (2023) 506, http://dx.doi.org/10.3390/math11030506.

[5] A. Sen, J.D. Baker, Q. Zhang, et al., Do more stringent policies reduce daily COVID-19 case counts? Evidence from Canadian provinces, Econ. Anal. Policy 78 (2023) 225–242, http://dx.doi.org/10.1016/j.eap.2023.03.006.

[6] B.F. Maier, D. Brockmann, Effective containment explains subexponential growth in recent confirmed COVID-19 cases in China, Science 368 (6492) (2020) 742–756, http://dx.doi.org/10.1126/science.abb4557.

[7] M.A. Khan, A. Atangana, E. Alzahrani, The dynamics of COVID-19 with quarantined and isolation, Adv. Difference Equ. 2020 (1) (2020) 425, http://dx.doi.org/10.1186/s13662-020-02882-9.

[8] L. Kalachev, E.L. Landguth, J. Graham, Revisiting classical SIR modelling in light of the COVID-19 pandemic, Infect. Dis. Model. 8 (1) (2023) 72–83, http://dx.doi.org/10.1016/j.idm.2022.12.002.

[9] F.P. Polack, S.J. Thomas, N. Kitchin, et al., Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine, N. Engl. J. Med. 383 (27) (2020) 2603–2615, http://dx.doi.org/10.1056/NEJMoa2034577.

[10] B. Yang, Z. Yu, Y. Cai, The impact of vaccination on the spread of COVID-19: Studying by a mathematical model, Physica A 590 (2022) 126717, http://dx.doi.org/10.1016/j.physa.2021.126717.

[11] G. Vivekanandhan, M.N. Zavareh, H. Natiq, F. Nazarimehr, K. Rajagopal, M. Svetec, Investigation of vaccination game approach in spreading covid-19 epidemic model with considering the birth and death rates, Chaos Solitons Fractals 163 (2022) 112565, http://dx.doi.org/10.1016/j.chaos.2022.112565.

[12] I. Cooper, A. Mondal, C.G. Antonopoulos, A SIR model assumption for the spread of COVID-19 in different communities, Chaos Solitons Fractals 139 (2020) 110057, http://dx.doi.org/10.1016/j.chaos.2020.110057.

[13] S. Ghosh, A. Senapati, A. Mishra, J. Chattopadhyay, D. Ghosh, Reservoir computing on epidemic spreading: A case study on COVID-19 cases, Phys. Rev. E 104 (2021) 014308, http://dx.doi.org/10.1103/PhysRevE.104.014308.

[14] B. Nikparvar, M.M. Rahman, F. Hatami, J.C. Thill, Spatio-temporal prediction of the COVID-19 pandemic in US counties: modeling with a deep LSTM neural network, Sci. Rep. 11 (2021) 201715, http://dx.doi.org/10.1038/s41598-021-01119-3.

[15] C. Murphy, E. Laurence, A. Allard, Deep learning of stochastic contagion dynamics on complex networks, Nature Commun. 12 (2021) 4720, http://dx.doi.org/10.1038/s41467-021-24732-2.

[16] S. Zhang, J. Ponce, Z. Zhang, G. Lin, G. Karniadakis, An integrated framework for building trustworthy data-driven epidemiological models: Application to the COVID-19 outbreak in New York City, PLoS Comput. Biol. 17 (9) (2021) e1009334, http://dx.doi.org/10.1371/journal.pcbi.1009334.

[17] P. Blonigan, J. Ray, C. Safta, Forecasting multi-wave epidemics through Bayesian inference, Arch. Comput. Methods Eng. 28 (6) (2021) 4169–4183, http://dx.doi.org/10.1007/s11831-021-09603-9.

[18] H.C. Yang, Y. Xue, Y. Pan, et al., Time fused coefficient SIR model with application to COVID-19 epidemic in the United States, J. Appl. Stat. 50 (11–12) (2021) 2373–2387, http://dx.doi.org/10.1080/02664763.2021.1936467.

[19] A. Bousquet, W.H. Conrad, S.O. Sadat, N. Vardanyan, Y. Hong, Deep learning forecasting using time-varying parameters of the SIRD model for Covid-19, Sci. Rep. 12 (2022) 3030, http://dx.doi.org/10.1038/s41598-022-06992-0.

[20] F. Amaral, W. Casaca, C.M. Oishi, J.A. Cuminato, Simulating immunization campaigns and vaccine protection against COVID-19 pandemic in Brazil, IEEE Access 9 (2021) 126011–126022, http://dx.doi.org/10.1109/ACCESS.2021.3112036.

[21] F. Amaral, W. Casaca, C.M. Oishi, Towards providing effective data-driven responses to predict the Covid-19 in Sao Paulo and Brazil, Sensors 21 (2) (2021) 540, http://dx.doi.org/10.3390/s21020540.

[22] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, J. Comput. Phys. 378 (2019) 686–707, http://dx.doi.org/10.1016/j.jcp.2018.10.045.

[23] E.O. Oluwasakin, A.Q.M. Khaliq, Data-driven deep learning neural networks for predicting the number of individuals infected by COVID-19 Omicron variant, Epidemiologia 4 (2023) 420–453, http://dx.doi.org/10.3390/epidemiologia4040037.

[24] World Health Organization, Coronavirus disease (COVID-19) Weekly Epidemiological Updates and Monthly Operational Updates, Available online: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/situation-reports.

[25] R.J. Boyton, D.M. Altmann, The immunology of asymptomatic SARS-CoV-2 infection: what are the key questions? Nat. Rev. Immunol. 21 (2021) 762–768, http://dx.doi.org/10.1038/s41577-021-00631-x.

[26] J.S. Jia, X. Lu, Y. Yuan, G. Xu, N.A. Christakis, Population flow drives spatio-temporal distribution of COVID-19 in China, Nature 582 (7812) (2020) 389–394, http://dx.doi.org/10.1038/s41586-020-2284-y.

[27] H.S. Badr, H. Du, M. Marshall, E. Dong, L.M. Gardner, Association between mobility patterns and COVID-19 transmission in the USA: a mathematical modelling study, Lancet Infect. Dis. 20 (11) (2020) 1247–1254, http://dx.doi.org/10.1016/S1473-3099(20)30553-3.

[28] Z. Rapti, J. Cuevas-Maraver, E. Kontou, S. Liu, et al., The role of mobility in the dynamics of the COVID-19 epidemic in Andalusia, Bull. Math. Biol. 85 (6) (2023) 54, http://dx.doi.org/10.1007/s11538-023-01152-5.

[29] M. Soheili, S. Khateri, F. Moradpour, et al., The efficacy and effectiveness of COVID-19 vaccines around the world: a mini-review and meta-analysis, Ann. Clin. Microbiol. Antimicrob. 22 (2023) 42, http://dx.doi.org/10.1186/s12941-023-00594-y.

[30] C. Rhee, M. Baker, V. Vaidya, et al., Incidence of nosocomial COVID-19 in patients hospitalized at a large US academic medical center, JAMA Netw. Open 3 (9) (2020) e2020498, http://dx.doi.org/10.1001/jamanetworkopen.2020.20498.

[31] U. Nations, Population and Vital Statistics Report, Available online: https://desapublications.un.org/publications/population-and-vital-statistics-report-volume-75.

[32] E. Mathieu, H. Ritchie, L. Rodés-Guirao, et al., Coronavirus Pandemic (COVID-19), Available online: https://ourworldindata.org/coronavirus.

[33] R. Li, S. Pei, B. Chen, et al., Substantial undocumented infection facilitates the rapid dissemination of novel coronavirus (SARS-CoV-2), Science 368 (6490) (2020) 489–493, http://dx.doi.org/10.1126/science.abb3221.

[34] W. Yu, Y. Guo, S. Zhang, et al., Proportion of asymptomatic infection and nonsevere disease caused by SARS-CoV-2 Omicron variant: A systematic review and analysis, J. Med. Virol. 94 (12) (2022) 5790–5801, http://dx.doi.org/10.1002/jmv.28066.

[35] N. Colburn, How long does it take for the COVID-19 vaccine to work?, Available online: https://wexnermedical.osu.edu/blog/how-long-for-covid-vaccine-to-work.

[36] Y. Wu, L. Kang, Z. Guo, et al., Incubation period of COVID-19 caused by unique SARS-CoV-2 strains: A systematic review and meta-analysis, JAMA Netw. Open 5 (8) (2022) e2228008, http://dx.doi.org/10.1001/jamanetworkopen.2022.28008.

[37] X. Hao, S. Cheng, D. Wu, et al., Reconstruction of the full transmission dynamics of COVID-19 in Wuhan, Nature 584 (7821) (2020) 420–424, http://dx.doi.org/10.1038/s41586-020-2554-8.

[38] C. Menni, A.M. Valdes, L. Polidori, et al., Symptom prevalence, duration, and risk of hospital admission in individuals infected with SARS-CoV-2 during periods of omicron and delta variant dominance: a prospective observational study from the ZOE COVID Study, Lancet 399 (10335) (2022) 1618–1624, http://dx.doi.org/10.1016/S0140-6736(22)00327-0.

[39] C. for Disease Control, Prevention, COVID-19 Pandemic Planning Scenarios, Available online: https://www.cdc.gov/coronavirus/2019-ncov/hcp/planning-scenarios.html.

[40] F.P. Polack, S.J. Thomas, N. Kitchin, et al., Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine, N. Engl. J. Med. 383 (2020) 2603–2615, http://dx.doi.org/10.1056/NEJMoa2034577.

[41] V.K.R. Chimmula, L. Zhang, Time series forecasting of COVID-19 transmission in Canada using LSTM networks, Chaos Solitons Fractals 135 (2020) 109864, http://dx.doi.org/10.1016/j.chaos.2020.109864.

[42] S. Triambak, D.P. Mahapatra, N. Mallick, et al., A new logistic growth model applied to COVID-19 fatality data, Epidemics 37 (2021) 100515, http://dx.doi.org/10.1016/j.epidem.2021.100515.

[43] World Health Organization, Classification of Omicron (B.1.1.529): SARS-CoV-2 Variant of Concern, Available online: https://www.who.int/news/item/26-11-2021-classification-of-omicron-(b.1.1.529)-sars-cov-2-variant-of-concern.